Appendix Contents

- Extended Analysis of Sensory Prompting
- Additional Qualitative Examples
- Evaluation on Additional Models and Datasets
- Visual Bias in Auditory Setting
- Extended Analysis of Sensory Axis Projections
- Extended Analysis of VQA in Text Space
- Full Prompted Generation Examples
- Visual Prompting Improves Perceptual Grounding in Image Generation

A Extended Analysis of Sensory Prompting

Additional instruction verbs. To assess the robustness of sensory prompting, we replicate the experiment using a broader set of $n = 10$ instruction verbs: *"conceptualize", "consider", "describe", "detail", "explain", "formulate", "imagine", "think (about)", "wonder", "write"*. We also include a null baseline where instead of an instructional sensory prompt, we prepend a random sentence drawn from captions of the DCI dataset. Results averaged across verbs (mean ± standard error) are shown in Figure 10.

Per-verb breakdown. In Figure 11 we provide alignment scores for each verb individually. Although overall trends are consistent, different verbs yield slightly different levels of alignment.

Potential for prompt optimization. The space of possible instruction prompts is vast and we have only explored a subset. For example, *"describe"* allows for higher alignment than *"imagine"*, as shown in Figure 12. Also see Figure 13 for the trend across Qwen3 sizes and generation lengths. This highlights the potential for optimizing prompts to maximize alignment. A current limitation, however, is that alignment evaluation is computationally expensive, making systematic prompt search challenging.

Instruction vs. non-instruction prompting. Finally, we compare instruction-style prompting (*"imagine what it would be like to see..."*) with analogous non-instructional forms (*"they imagined seeing..."*). As shown in Figure 14, alignment is much higher under the instruction form, indicating that instruction-tuned models are actively following the command rather than responding to the semantic content of the word itself.

Overgeneration reduces alignment. While increasing generation length up to 256 tokens improves sensory alignment, we observe that performance can decline again at 512 tokens (Figure 15). This suggests that overly long generations may lead to semantic drift or off-topic elaboration. See Appendix G.4 for qualitative examples.

Layer-wise evaluation. One concern is that the observed benefits of sensory prompting might reflect only *superficial priming* of the final layer: in other words, that the LLM is simply adjusting its last-step representation so the output head is biased toward modality-related words (e.g., "seeing" toward visual descriptors, "hearing" toward auditory ones), rather than inducing a deeper change in its internal representations. To test this, we compute alignment scores layer-by-layer rather than only on the mean-layer embedding.

Figure 16shows layer-wise results for Qwen3 evaluated on both the WiT (image–text) and AudioCaps (audio–text) datasets. The sensory prompting trend is preserved across layers, indicating that the effect is not confined to superficial bias at the output layer but instead reflects a consistent shift in the model's intermediate representations. Interestingly, we also find that using the mean embedding across all layers often yields higher alignment than any single layer. One possible explanation is that averaging smooths out layer-specific noise while retaining complementary information across the hierarchy, though a full understanding of this phenomenon remains

open for future study. Figure 16 shows layer-wise results for Qwen3, demonstrating that averaging captures consistent trends across layers.

Redirecting sensory cues. We generate 128-token outputs from Qwen3-32B using either a SEE or HEAR cue, then rewrite them with redirection templates (Figure 17b) that flip the modality. This produces a clear double dissociation: generations that redirect the cue from SEE to HEAR align more with audio encoders and less with vision, while generations that redirect the cue from HEAR to SEE shift toward vision encoders and away from audio (Figure 17a).

Note that redirection remains effective even when the initial output is lossy—for example, a visual caption prompted with HEAR may drop visual detail and introduce auditory description. Yet rewriting back to SEE restores alignment, suggesting that the model can correctly make cross-modal inferences (e.g., reconstructing visual structure from auditory framing). These results show that sensory prompts causally steer representations toward or away from modality-specific subspaces. Examples of redirected generations are provided in Appendix G.2.

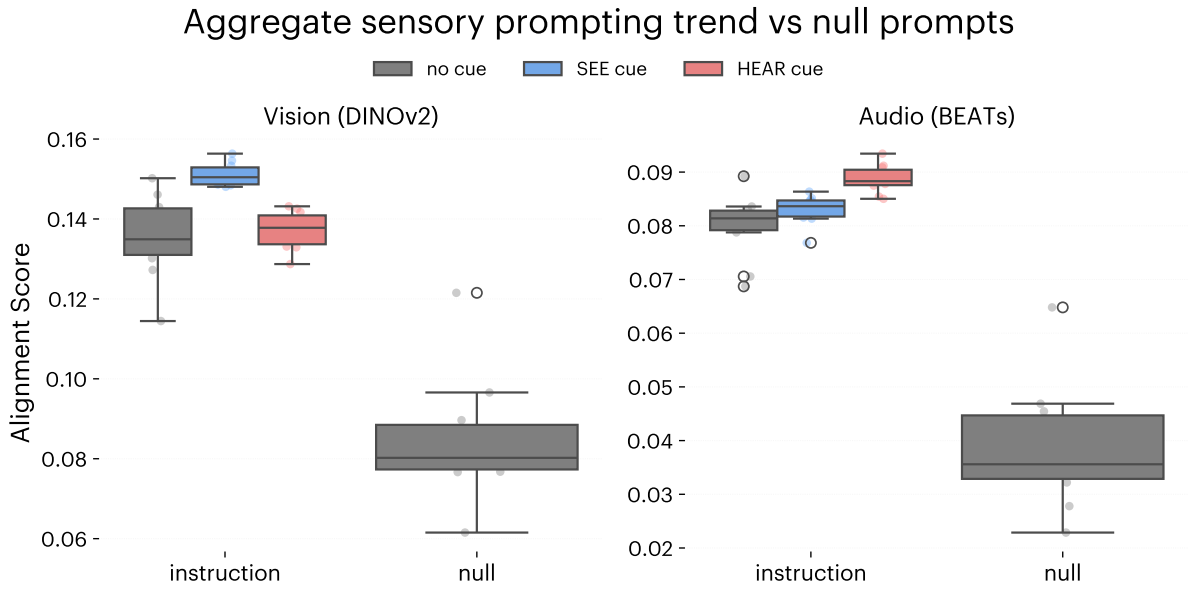

Figure 10: Extension of Figure 3 to additional verbs and null prompts (sentences drawn from DCI captions) for prompting.

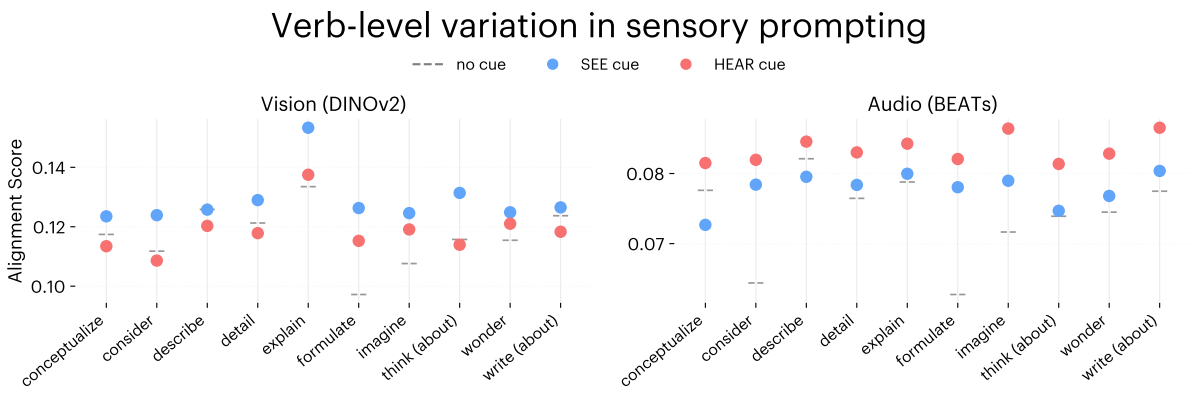

Figure 11: Extension of Figure 3 to additional verbs for prompting.

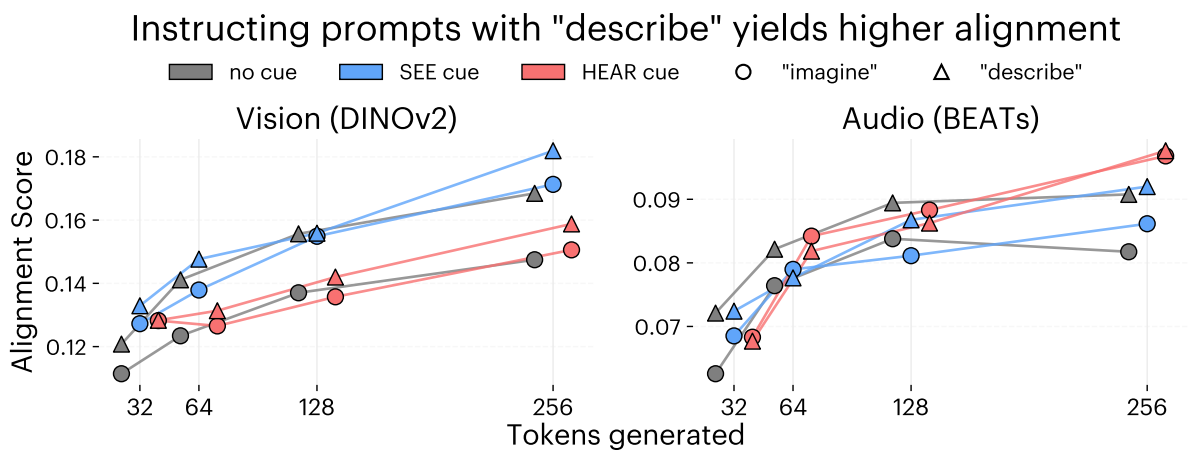

Figure 12: Instructing the LLM with “describe” can yield better alignment than “imagine”.